

CDS 101 -Final Project Report

Exponentially Decaying / Nivia, Neel, Justin

2026-05-13

1. Problem Definition

- Our project investigates whether weather conditions are related to equine incidents at New York race tracks. The main question we are studying is: How significant is the relationship between equine incidents and weather conditions?
- This topic is important because equine deaths and injuries are serious safety concerns in horse racing. By studying incident patterns over time and comparing them with weather conditions, we may be able to identify whether certain weather factors are associated with higher or lower incident counts.
- The objective of this analysis is to explore trends in equine incidents from 2009 to 2025, compare incidents across different years, tracks, incident types, seasons, and weather categories, and determine whether weather appears to have a noticeable relationship with incident patterns.
- A key assumption is that the incident reports are accurate and consistently recorded. Another assumption is that the weather information connected to each incident is reliable enough to be grouped into categories such as clear, rainy, cloudy, snowy, or unknown.

2. Data Acquisition & Description

- The name of the dataset is Equine Death and Breakdown from the data.gov website which is an official website of the United States government.
- Obtained through a download link from data.gov website.
- There are a lot of variables / features such as Incident Type, Horse, Jockey Driver, Track, and so forth but we will be focusing primarily on two variables, Death or Injury and Weather Conditions. Death or Injury variable shows if a horse died in an incident or was just injured. Weather Conditions variable shows the weather conditions at the time of the incident. We will also be using “Track”, “Year”, and “Incident Date”, and “Incident Type” columns to help supplement our findings.
- The dataset has 5,383 rows and 13 columns.

3. Data Cleaning & Preprocessing

- This dataset was extremely messy and we had to do a lot of cleaning to make it usable and went from 5,383 rows to just 1,702. We got rid of any rows that had NA values in the Weather Condition and Death and Injury variables. The incident date variable had to be changed to a date type. The column names were fixed to make it easier to use and read. Originally the dataset's weather condition variable had 1000+ different unique entries but we wanted them to be categorized in just a few simple ones. The weather_clean variable that was created that holds the weather_conditions values without all the unnecessary information and spaces. This is then piped into weather_group which categorizes the specific weather condition happening at the time of the incident such as "Rain", "Snow", or "Clear" and so forth. The death_or_injury_clean variable cleans up the 10 unique values and categorizes them into 2 simple ones "death" or "injury." After creating these new variables we filtered them to only include the values that were relevant to our research question. A season variable was also created for EDA. Finally, the Death and Injury variable was turned to binary 0/1 variable since its a categorical variable and we need a numerical values to make a linear regression model.

```
#Fix column names
horses_clean <- horses %>%
  rename(year = Year, incident_date = `Incident Date`, track = Track,
         inv_location = `Inv Location`,
         racing_type_description = `Racing Type Description`,
         division = Division, weather_conditions = `Weather Conditions`,
         horse = Horse, trainer = Trainer, jockey_driver = `Jockey Driver`,
         incident_description = `Incident Description`,
         death_or_injury = `Death or Injury`,
         incident_type = `Incident Type`)

#Fix dates
horses_clean$incident_date <- mdy(horses_clean$incident_date)

#Get rid of NA entries
horses_clean <- horses_clean %>%
  filter(!is.na(weather_conditions), !is.na(death_or_injury))

#Fix weather_condition column
#Used ChatGPT to help with this cleaning part
horses_clean <- horses_clean %>%
  mutate(
    weather_clean = weather_conditions %>%
      str_to_lower() %>%
      str_trim() %>%
      str_replace_all("[0-9]+", "") %>%
      str_replace_all("[^a-z ]", "") %>%
      str_squish()
  ) %>%
```

```

mutate(
  weather_group = case_when(
    str_detect(weather_clean, "rain|shower|drizzle|raine") ~ "Rain",
    str_detect(weather_clean, "snow|flurr|sleet") ~ "Snow",
    str_detect(weather_clean, "thunder|storm") ~ "Storm",
    str_detect(weather_clean, "cloud|coudy|cloudy|clody") ~ "Cloudy",
    str_detect(weather_clean, "clear|sun|fair|dry") ~ "Clear",
    str_detect(weather_clean, "windy|fog|mist|hazy|wind") ~ "Other Weather",
    weather_clean %in% c("", "na", "n a", "unc") | is.na(weather_clean) ~ "Unknown",
    TRUE ~ "Other"
  )
)

#Clean death_or_injury column
horses_clean <- horses_clean %>%
  mutate(
    death_or_injury_clean = death_or_injury %>%
      str_to_lower() %>%
      str_trim(),

    death_or_injury_clean = case_when(
      str_detect(death_or_injury_clean, "death|dead|euthanasia") ~ "Death",
      str_detect(death_or_injury_clean, "lame|injury|lameness") ~ "Injury",
      TRUE ~ "Other"
    )
  )

horses_clean <- horses_clean %>%
  filter(death_or_injury_clean != "Other") %>%
  select(year, incident_date,
    incident_type,
    track,
    incident_description,
    weather_group,
    death_or_injury_clean) %>%
  filter(weather_group != "Other" & weather_group != "Unknown")

#Create season variable (Code reference linked below)
horses_clean <- horses_clean %>%
  mutate(
    season = case_when(
      month(incident_date) %in% 9:11 ~ "Fall",
      month(incident_date) %in% c(12,1,2) ~ "Winter",
      month(incident_date) %in% 3:5 ~ "Spring",
      TRUE ~ "Summer"
    )
  )

```

4. Exploratory Data Analysis (EDA)

```
summary(horses_clean)
```

```
##      year      incident_date      incident_type      track
## Min.   :2009   Min.   :2009-04-28   Length:1702      Length:1702
## 1st Qu.:2012   1st Qu.:2012-01-20   Class :character   Class :character
## Median :2016   Median :2016-10-01   Mode  :character   Mode  :character
## Mean   :2017   Mean   :2017-03-27
## 3rd Qu.:2022   3rd Qu.:2022-09-15
## Max.   :2026   Max.   :2026-04-12
## incident_description weather_group      death_or_injury_clean
## Length:1702          Length:1702          Length:1702
## Class :character      Class :character      Class :character
## Mode  :character      Mode  :character      Mode  :character
##
##
##
##      season
## Length:1702
## Class :character
## Mode  :character
##
##
##
```

i.

```
library(ggplot2)
ggplot(horses, aes(x = Year, color = 'white')) +
  geom_line(stat = "count") +
  geom_point(stat = "count", size = 3) +
  labs(
    x = "Year",
    y = "# of Incidents",
    title = "Equine Incidents Over Time"
  ) +
  theme_minimal()
```

- The data overall shows a clear decline in equine incidents from 2009 to 2026 with few inconsistencies. Incident counts start at a moderate level in the late 2000s and rise very fast and peak in 2012. After that the count steadily falls. This pattern equine incidents have generally decreased over the years. The inconsistencies show how the incident counts do not decrease perfectly smoothly, indicating that many external factors such as heat, cold, rain, etc. contribute to short term increases.

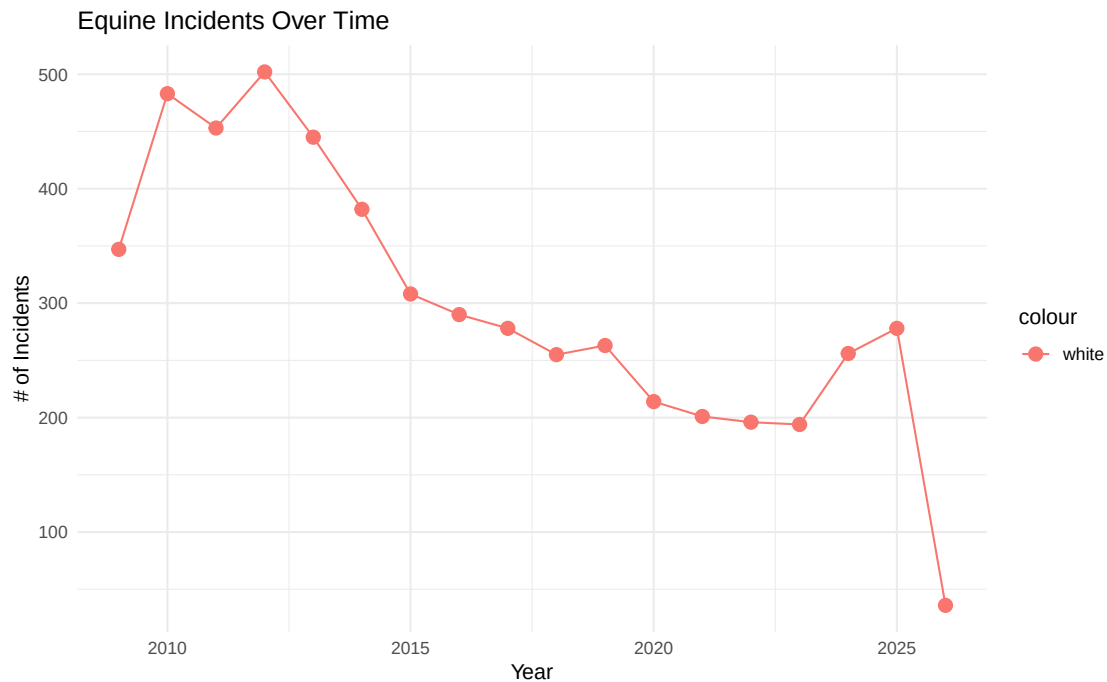
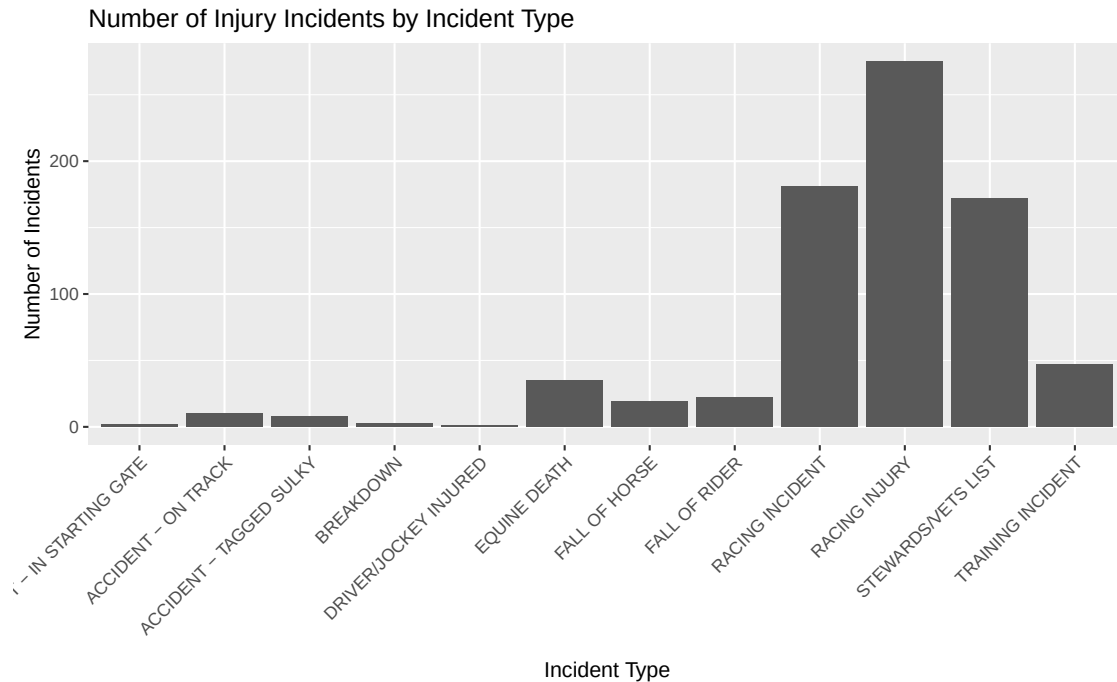


Figure 1: Relationship between time of year and weather related equine incidents

ii.

```
horses_clean %>%
  filter(death_or_injury_clean == "Injury") %>%
  ggplot(aes(x = incident_type)) +
  geom_bar() +
  labs(
    title = "Number of Injury Incidents by Incident Type",
    x = "Incident Type",
    y = "Number of Incidents"
  ) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



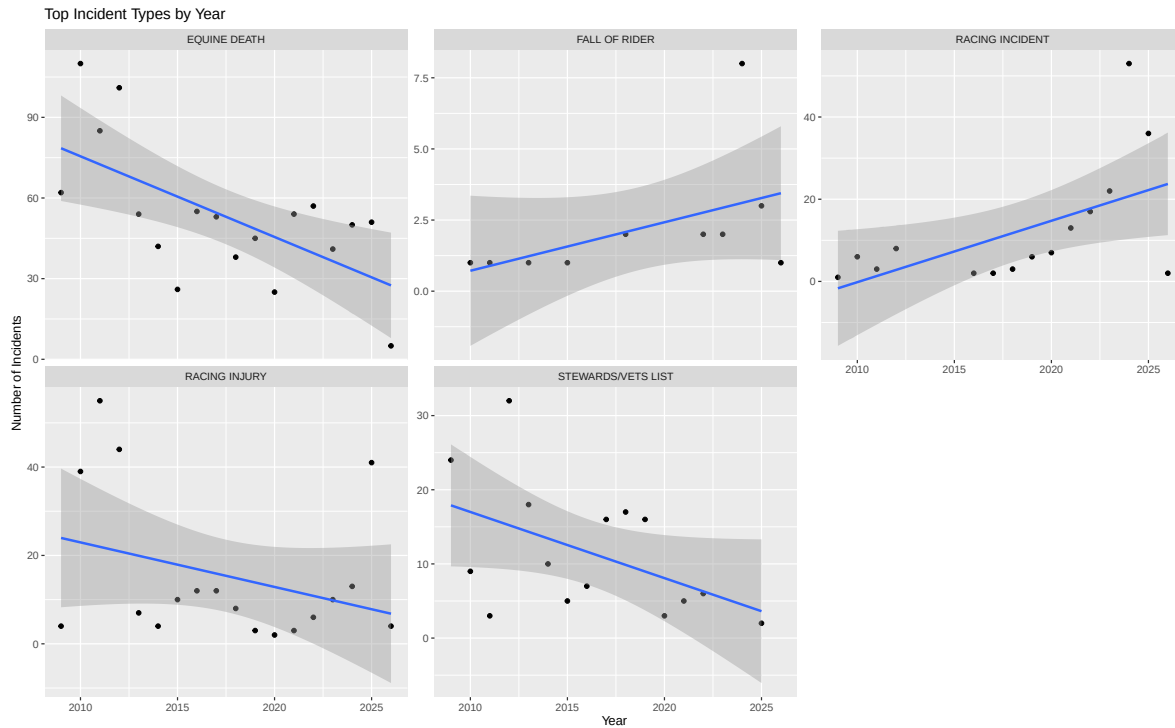
- This graph connects to our research question by showing which types of equine incidents happen most often in the data set. Since our project is studying the relationship between weather related variables and equine deaths or breakdowns, we first need to understand what kinds of incidents are being recorded. The most common incident types are the main categories we can later compare with weather conditions, temperature, precipitation, or season. This graph does not show a weather relationship by itself, but it gives important background for the weather analysis.

iii.

```
horses_clean %>%
  filter(incident_type %in% c(
    "EQUINE DEATH",
    "STEWARDS/VETS LIST",
    "RACING INCIDENT",
    "RACING INJURY",
    "FALL OF RIDER"
  )) %>%
  count(year, incident_type) %>%
  ggplot(aes(x = year, y = n)) +
  geom_point() +
  geom_smooth(method = "lm") +
  facet_wrap(~ incident_type, scales = "free_y") +
  labs(
    title = "Top Incident Types by Year",
    x = "Year",
```

```
y = "Number of Incidents"
)
```

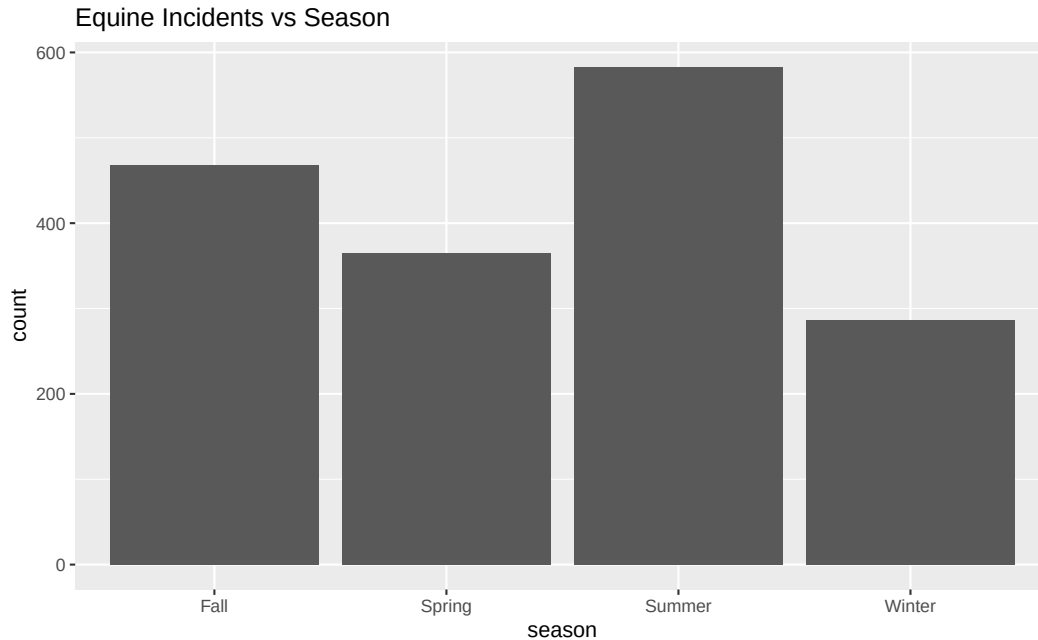
```
## 'geom_smooth()' using formula = 'y ~ x'
```



- These graphs help connect to our question by showing how incident types changed over time. If certain years had higher numbers of specific incident categories, those changes could later be compared with weather patterns from those same years. This helps us look for possible trends before testing whether weather-related variables are connected to equine deaths or breakdowns. The graph does not prove that weather caused the incidents, but it helps identify time-based patterns that may be useful for the rest of the project.

iv.

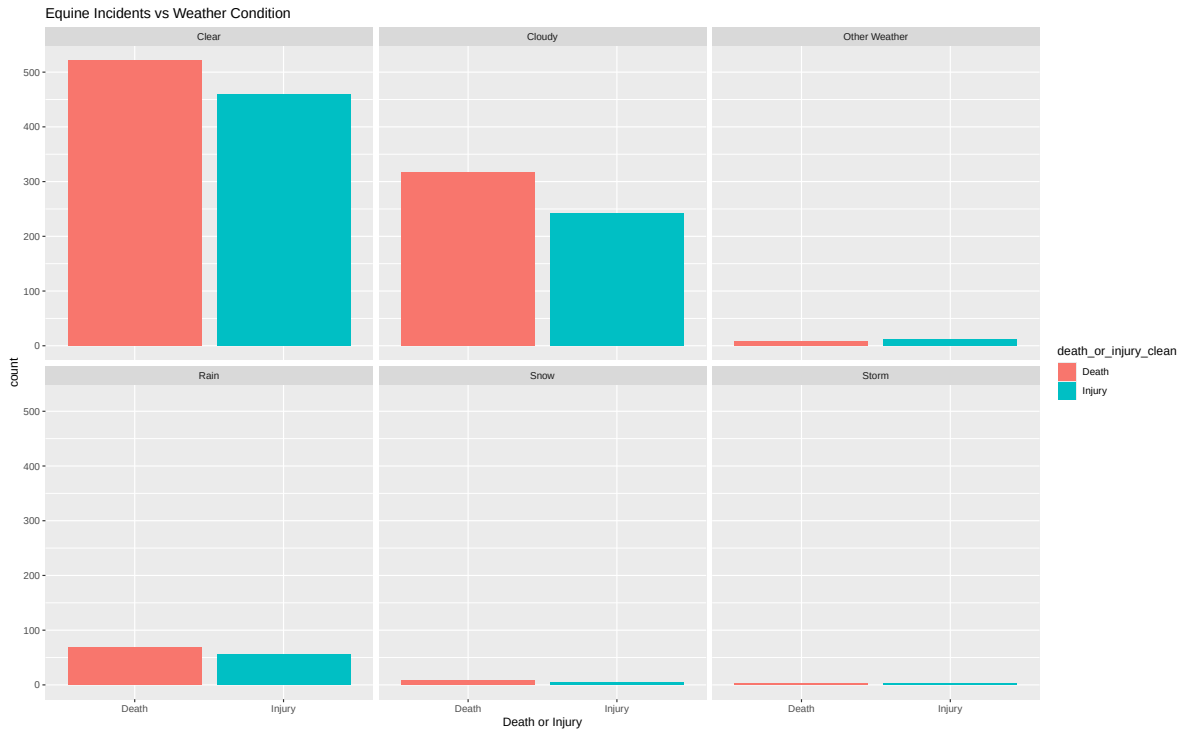
```
horses_clean %>%
  ggplot(aes(x = season)) +
  geom_bar() +
  labs(
    title = "Equine Incidents vs Season"
  )
```



- We assume that Winter has the most inclement weather out of all the seasons and yet we see that Winter has the least amount of equine deaths and injuries which could indicate a weak correlation between season and equine incidences if we take our assumption as the truth. Summer season has the most incidents but that could be due to there being more races compared to other seasons.

vi.

```
horses_clean %>%
  ggplot(aes(x = death_or_injury_clean, fill = death_or_injury_clean)) +
    geom_bar() +
    facet_wrap(~ weather_group) +
    labs(
      x = "Death or Injury",
      title = "Equine Incidents vs Weather Condition")
```

- From observing the graph there does not appear to be a clear correlation between inclement weather condition and equine death or injury. The weather category with the most equine deaths and incidents is the clear weather with cloudy weather being the second most. If there was a strong correlation between inclement weather condition and equine we would expect to see rain, snow, storm, and other weather categories have higher counts compared to clear.

5. Visualization Quality and Storytelling

- For the incident type section, we used a bar graph first because we are looking at incident type which is categorical. This graph makes sense because it lets us compare how many incidents happened in each incident type, and we flipped the graph so the incident type names are easier to read instead of crowded on the x-axis. For the second graph, we used a line graph because we are looking at incident type by year, which helps show how the incident categories changed over time instead of only showing the total number of incidents.
- For graph iv. we used a bar graph because we were comparing two categorical variables to each other and we wanted to see which season had the most equine incidents. The x-axis is the seasons variable and the y-axis is the amount of incidents. To make the plot more visible we needed to adjust some of the parameters.
- For graph vi. we used faceted bar graphs because we were comparing multiple different categorical categories with equine incidents to see which category had the most incidents. The different colors help visualize death versus injury with respect to the specific weather condition. For visibility we also had to make the graph bigger.

6. Modeling Approach

Explain how you framed the problem and which models you chose:

- Type of task (regression, classification, etc.).
- Baseline model or heuristic, if used.
- Main model(s) chosen and why they are appropriate.

For our modeling approach, we followed the plan from our proposal by using ideas from the blood pressure assignment and the car prices assignment. In the blood pressure assignment, we looked at relationships between variables and used linear models to understand trends. In the car prices assignment, we used a model with more than one explanatory variable. For our project, we used the number of incidents as the response variable because incident count is numeric and works with a linear model. The explanatory variables we focused on were year, season, weather condition, and incident type. These variables match the graphs we made because we looked at incidents over time, incident categories, incidents by season, and incidents by weather condition.

For our baseline model, we would use a simple heuristic, which is a simple rule for making a prediction. Our heuristic would predict the average number of incidents every time without using year, season, weather condition, or incident type. This gives us a simple comparison for our multi-variable linear model. If the multi-variable model makes better predictions than just using the average incident count, then that would suggest that the variables we chose are useful for explaining patterns in the data.

This connects to our research question because we are trying to see whether weather-related variables are connected to equine deaths and breakdowns. The incidents-over-time graph shows that incident counts changed by year, so year is an important variable to include. The incident type graphs show which categories happened the most and whether those categories changed over time. The season and weather condition graphs connect more directly to the weather part of our project because they show how incidents are spread across different seasons and weather conditions. Using a multi-variable linear model helps us look at these variables together instead of only looking at one graph or one factor at a time.

7. Model Implementation & Evaluation

8. Conclusions & Recommendations

- In conclusion, our project helped us see that equine deaths and breakdowns are affected by more than one factor. We looked at incident types, time trends, seasons, and weather conditions to see what patterns showed up in the data. The graphs showed that some incident categories happened more often than others, and the number of incidents was not the same every year or every season. Even though weather was the main focus of our research question, the results show that we have to be careful when interpreting the graphs. Higher incident counts in one weather condition do not automatically mean that weather caused

more incidents. It may also depend on a vast number of reasons due to the various parts of racing.

- Some limitations of our research could be the phenomena known as “survivorship bias” similar to lab 11 we did in CDS 102. The apparent relationship we are seeing between equine deaths and inclement weather is affected by the fact we don’t get data on how many races were canceled due to inclement weather being too bad to race. So we can’t say that weather has no correlation to equine incidents because we have to consider survivorship bias and how many horses could have been injured if they did race. If we wanted better representation we would need to add those canceled races to the dataset and observe the results.
- A useful extension for our project would be to expand our data and add more detailed information about the horses themselves. For example, we could determine the age of the horse as well as the condition to determine if the horse had any previous injuries that would lower their chances at survival. Another important extension would be to include data on if races were cancelled due to extreme weather, as we cannot see how many deaths would have been caused by weather if the races were actually done.

9. Code Quality & Reproducibility

- Someone can reproduce our results by opening the project in RStudio and running the final_report_template.Rmd file from top to bottom. The horses.csv file needs to be in the project folder so the dataset loads correctly. The required packages are tidyverse, readr, ggplot2, broom, modelr, dplyr, lubridate, and reader. Our code is organized in the same order as the report: data loading, data cleaning, EDA graphs, modeling, and evaluation. The cleaned dataset horses_clean is used for the graphs and model, so the cleaning code has to run first.
- GitHub repository

```
## R version 4.5.2 (2025-10-31 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=English_United States.utf8
## [2] LC_CTYPE=English_United States.utf8
## [3] LC_MONETARY=English_United States.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United States.utf8
##
## time zone: America/New_York
## tzcode source: internal
##
```

```
## attached base packages:
## [1] stats      graphics  grDevices utils      datasets  methods   base
##
## other attached packages:
## [1] modelr_0.1.11  broom_1.0.12  forcats_1.0.1  purrr_1.2.1
## [5] readr_2.1.6    tidyr_1.3.2   tibble_3.3.1   tidyverse_2.0.0
## [9] ggplot2_4.0.2  stringr_1.6.0  lubridate_1.9.5 dplyr_1.2.0
##
## loaded via a namespace (and not attached):
## [1] generics_0.1.4    lattice_0.22-7  stringi_1.8.7    hms_1.1.4
## [5] digest_0.6.39     magrittr_2.0.4  evaluate_1.0.5   grid_4.5.2
## [9] timechange_0.4.0  RColorBrewer_1.1-3 fastmap_1.2.0    Matrix_1.7-4
## [13] backports_1.5.0   mgcv_1.9-3      scales_1.4.0     cli_3.6.5
## [17] rlang_1.1.7       crayon_1.5.3    splines_4.5.2    bit64_4.6.0-1
## [21] withr_3.0.2       yaml_2.3.12     otel_0.2.0       tools_4.5.2
## [25] parallel_4.5.2    tzdb_0.5.0      vctrs_0.7.1      R6_2.6.1
## [29] lifecycle_1.0.5   bit_4.6.0       vroom_1.7.0      pkgconfig_2.0.3
## [33] pillar_1.11.1     gtable_0.3.6    glue_1.8.0       xfun_0.56
## [37] tidyselect_1.2.1  rstudioapi_0.18.0 knitr_1.51       farver_2.1.2
## [41] nlme_3.1-168      htmltools_0.5.9  rmarkdown_2.30   labeling_0.4.3
## [45] compiler_4.5.2    S7_0.2.1
```

10. References

- CDS Textbook
- Equine Breakdown, Death, Injury and Incident Database Overview
- NYS Gaming Commision Equine Breakdown
- Dates to seasons column tutorial